Ex. No. 1	WORKING WITH WIDGETS AND LAYOUTS
Date of Exercise	11.08.2026
Git Hub Host Link	

Aim

To develop a Flutter application using basic widgets and layout widgets such as Container, Row, Column, Expanded, Card, and List View to create a simple student profile/dashboard interface.

Description

Flutter provides various widgets to design user interfaces. In this experiment, basic widgets such as Text, Icon, Container, Row, Column, Card, List Tile, and List View are used to create a student profile application. The application demonstrates how widgets can be arranged using different layout structures.

Procedure:

1. Open **VS Code** and launch the Terminal.
2. Check whether Flutter is installed correctly using:

   ```
   flutter doctor
   ```

3. Check the available devices using:

   ```
   flutter devices
   ```

4. Create a new Flutter project using:

   ```
   flutter create widget_layout
   ```

5. Move into the project directory:

   ```
   cd widget_layout
   ```

6. Open the project in VS Code:

   ```
   code .
   ```

7. Run the default Flutter application:

   ```
   flutter run
   ```

8. Open the **lib/main.dart** file and delete the existing code.
9. Import the Flutter Material package and create the basic application using MaterialApp, Scaffold, AppBar, and Text.
10. Replace the body with a **Column** widget and add student information such as student name, department, and laboratory name.
11. Add a **Container** widget with padding and margin to display the student profile.
12. Add a **Row** widget with different icons such as Home, Person, and Settings using MainAxisAlignment.space Evenly.
13. Add **Card** widgets with ListTile to display information such as Flutter and Mobile Application Development.
14. Add a **ListView** containing multiple ListTile items such as:
    - Flutter
    - Cloud Computing
    - Programming
15. Save the program and run the application using:

```
flutter run
```

16. Observe the student profile/dashboard interface displayed on the selected device.

Program

```
import 'package:flutter/material.dart';

void main() {

 runApp(const MyApp());

}

class MyApp extends StatelessWidget {

  const MyApp({super.key});

  @override

  Widget build(BuildContext context) {

    return MaterialApp(

    debugShowCheckedModeBanner: false,

    title: 'Student Profile',

      theme: ThemeData(
```

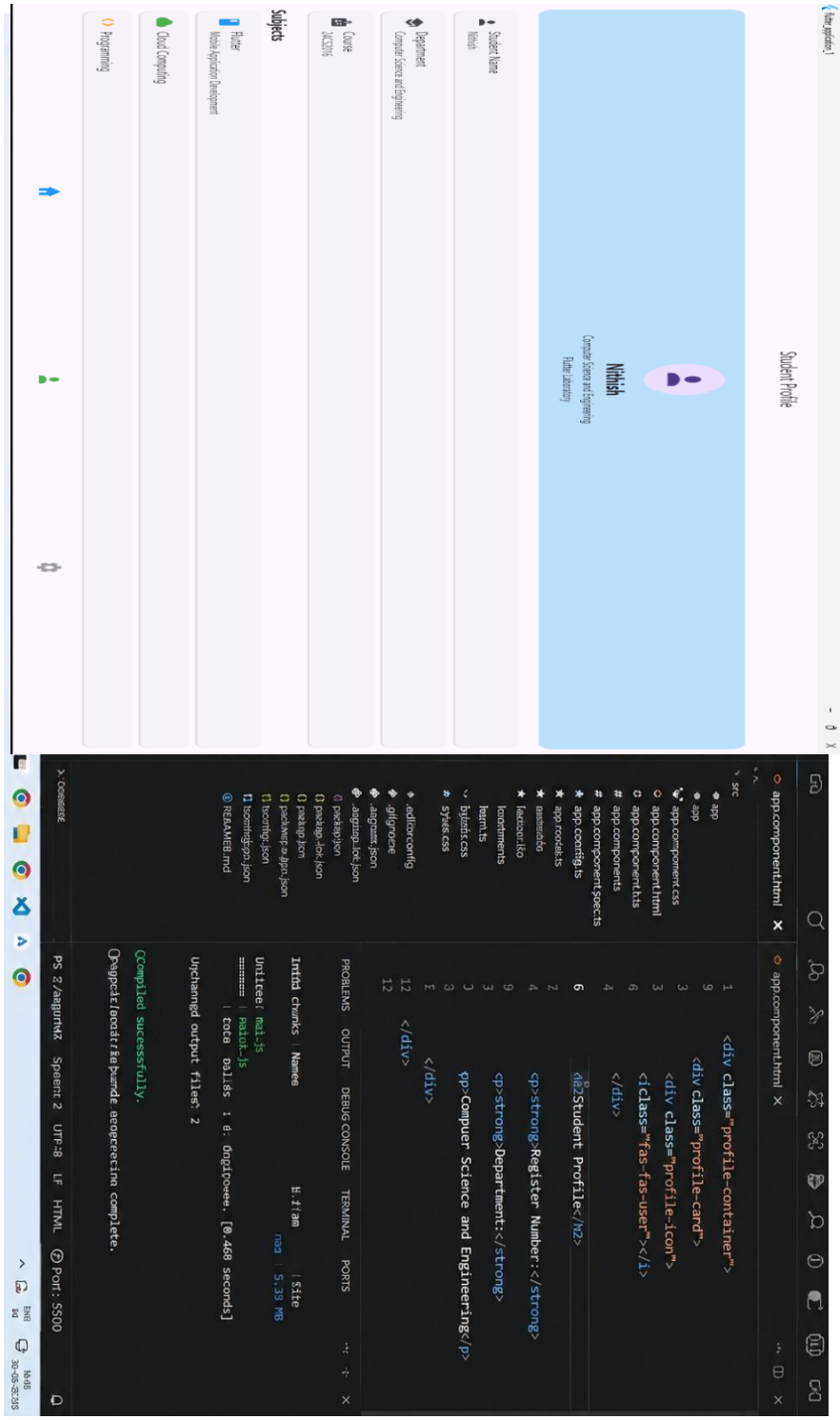
```
      colorScheme: ColorScheme.fromSeed(seedColor: Colors.blue),
      useMaterial3: true,
    ),
    home: const StudentProfile(),
   );
  }
}
class StudentProfile extends StatelessWidget {
  const StudentProfile({super.key});
  @override
  Widget build(BuildContext context) {
   return Scaffold(
    appBar: AppBar(
      title: const Text('Student Profile'),
      backgroundColor: Colors.blue,
      foregroundColor: Colors.white,
    ),
    body: ListView(
      padding: const EdgeInsets.all(16),
      children: [
       Container(
        padding: const EdgeInsets.all(20),
        margin: const EdgeInsets.only(bottom: 15),
        decoration: BoxDecoration(
         color: Colors.blue.shade50,
         borderRadius: BorderRadius.circular(12),
        ),
```

```
child: const Column(
  children: [
    Icon(
      Icons.person,
      size: 70,
      color: Colors.blue,
    ),
    SizedBox(height: 10),
    Text(
      'Student Profile',
      style: TextStyle(
      fontSize: 24,
        fontWeight: FontWeight.bold,
      ),
    ),
    SizedBox(height: 5),
    Text(
    'URK23CS8002',
      style: TextStyle(fontSize: 18),
    ),
    Text(
      'Computer Science and Engineering',
      textAlign: TextAlign.center,
    ),
  ],
),
```

```
      ),
      const Text(
        'Quick Access',
        style: TextStyle(
        fontSize: 20,
          fontWeight: FontWeight.bold,
        ),
      ),
      const SizedBox(height: 10),
      Row(
        mainAxisAlignment: MainAxisAlignment.spaceEvenly,
        children: const [
          Icon(Icons.home, size: 35, color: Colors.blue),
          Icon(Icons.person, size: 35, color: Colors.green),
          Icon(Icons.settings, size: 35, color: Colors.orange),
        ],
      ),
      const SizedBox(height: 20),
      const Card(
        child: ListTile(
          leading: Icon(Icons.book, color: Colors.blue),
          title: Text('Flutter'),
          subtitle: Text(
            'Mobile Application Development',
          ),
        ),
      ),
```

```
        const Card(
          child: ListTile(
            leading: Icon(Icons.cloud, color: Colors.blue),
            title: Text('Cloud Computing'),
          ),
        ),
        const Card(
          child: ListTile(
            leading: Icon(Icons.code, color: Colors.blue),
            title: Text('Programming'),
          ),
        ),
      ],
    ),
  );
 }
}
```

Output


Student Profile
Nithish
Computer Science and Engineering
Flutter Laboratory
Student Name
Nithish
Department
Computer Science and Engineering
Course
JAVSCRIPT
Subjects
Flutter
Mobile Application Development
Cloud Computing
Programming
app.component.html

<h2>Student Profile</h2>
Register Number:
Department:
Computer Science and Engineering

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Compiled successfully.
PS Z:\anguntu> Speent 2 UTF-8 LF HTML Port: 5500

Result

Thus, a Student Profile/Dashboard application was successfully developed using Flutter with Container, Row, Column, Card, List View, and other basic widgets. The application was executed successfully and the expected user interface was displayed.